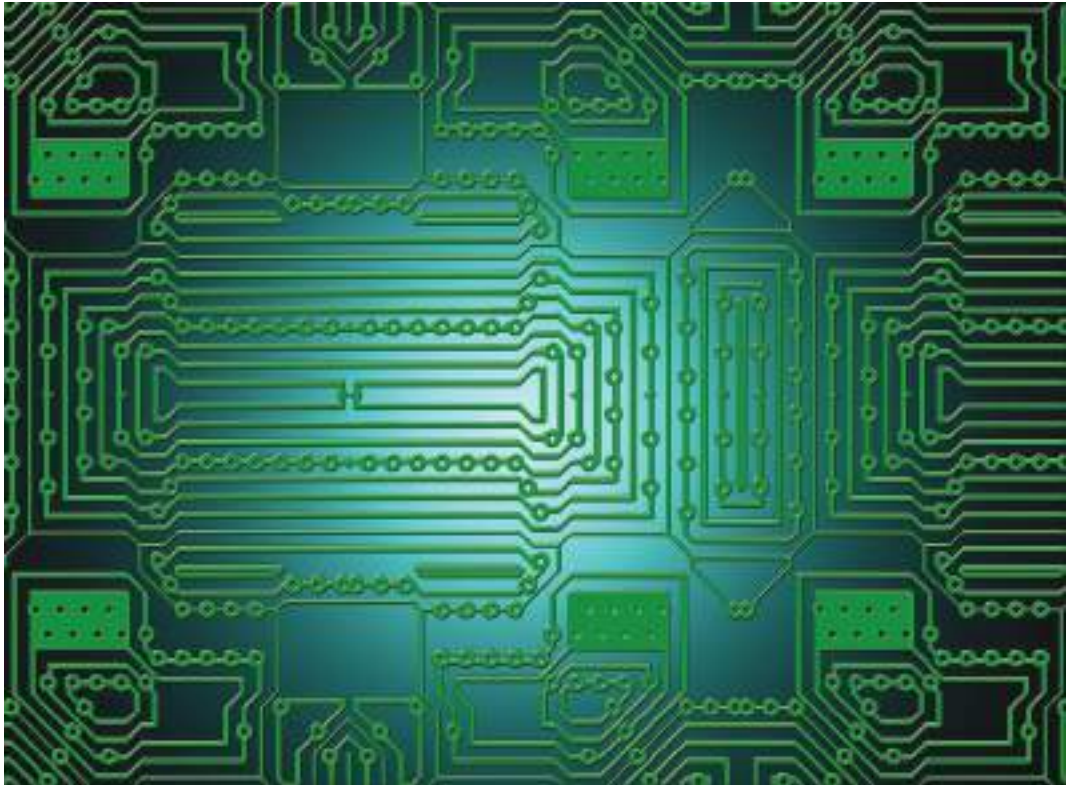


Chapter Thirteen

Radioactivity and Electronics



Radioactivity is one of the new topics that have flourished after the emergence of modern physics at the beginning of twentieth century. Radio activity is described in this chapter.

Electronics is the main drive of our present civilization- this statement is not exaggerated at all. The contribution of physics behind the present electronics has been discussed in this chapter, and the instruments that have changed our lives completely have been introduced .



By the end of this chapter, we will be able to-

- explain radioactivity.
- explain the characteristics of alpha, beta and gamma rays.
- describe the sequential development of electronics.
- differentiate between analogue and digital electronics.
- explain semiconductors and integrated circuits.

13.1 Radioactivity

We all know, there is a nucleus at the center of an atom, and there are protons and neutrons in the nucleus. The number of revolving electrons around the nucleus is equal to the number of protons in it. The radius of the nucleus is about one lakh times smaller than the radius of the atom. But the actual mass of the atom is the mass of its nucleus even though its size is small, because the mass of the electron is 1836 times less than that of the proton or neutron.

A proton is positively charged, so the nucleus cannot be formed only with protons because the protons will be thrown off due to strong repulsion force. So there is charge-less neutrons with protons in the nucleus, and nuclei become stable due to very strong nuclear force between neutrons and protons. Though there is only one proton at the center of a normal hydrogen atom, there are also hydrogen nuclei with one or two neutrons.

With the increase of the number of protons, to keep this nucleus stable, the number of neutrons also increases. But when the proton number exceeds 82, the nucleus starts to become unstable. But, no stable isotope is found for Technetium (43) and Promethium (61). These unstable nuclei try to become stable by emitting a kind of radiation. We call this process radioactivity. The radiation that comes out from the nucleus is called a radioactive ray.

It is not the fact that when proton number exceeds 82 (atomic number more than 82) only then the nucleus becomes radioactive. Any nucleus of other atoms may be radioactive too. We have classified atoms on the basis of their electron number which is equal to the number of protons. External properties, nature, chemical properties; every feature of an element depends on the classification of electrons. So the number of neutrons of an atom can be different even if the number of electrons and protons is equal. These types of elements, with different numbers of neutrons, are called isotopes of that element. So, an isotope of an element may be stable, but another isotope of that element may be unstable or radioactive. For example, we can say that element carbon in whose nucleus there are six protons and mainly three isotope. They are:

C_{12} : 6 protons and 6 neutrons

C_{13} : 6 protons and 7 neutrons

C_{14} : 6 protons and 8 neutrons

Among these three isotopes, C_{14} is unstable or radioactive.

In 1896, Henry Becquerel discovered the existence of radioactive rays from Uranium for the first time. Later on, Ernest Rutherford, Pierre Curie, Marie Curie and other scientists discovered radioactivity of other elements. Since radioactivity it is not influenced by external pressure, heat, electric or magnetic field, and cannot be controlled, it is accepted as a nuclear phenomenon. Due to radioactivity, radioactive rays emerge and this changes the structure of the nucleus and turn it into a different element, this is also observed.

The main three radioactive rays that emit from the nucleus are alpha, beta and gamma rays (Figure 13.01).

13.1.1 Alpha ray

Alpha ray or alpha particle is actually a helium nucleus. A helium nucleus contains two protons and two neutrons, so it is a charged particle. So its path of motion can be influenced by electric and magnetic field. When an alpha particle emits from a nucleus, it possesses some MeV of energy, so when it passes through air and collides with the molecule -atoms it can ionize them intensively. In air, the path of motion of an alpha particle is straight. Since an alpha particle is a helium nucleus, it cannot travel far through matter. It loses its energy by ionizing air molecule-atoms and travels approximately 6 cm through air before stopping. An alpha particle can be stopped by using a sheet of paper. It creates phosphorescence on zinc sulfide screen. An alpha particle creates many free electrons by ionizing molecule-atoms of air in its path of motion. Its presence can be detected or its energy can be measured easily by collecting these free electrons.

When a alpha particle comes out from a larger radioactive nucleus then this nucleus's atomic number will be reduced by two and its nucleon number will be reduced by four, because an alpha particle consists of two protons and two

neutrons. For example: an isotope of uranium turns into thorium by emitting an alpha particle.



The atomic number of uranium is 92 and thorium is 90. It is notable that number of electrons of an atom is not significant. Radioactive nuclei of atoms can easily give up or receive extra electrons to and from its surroundings.

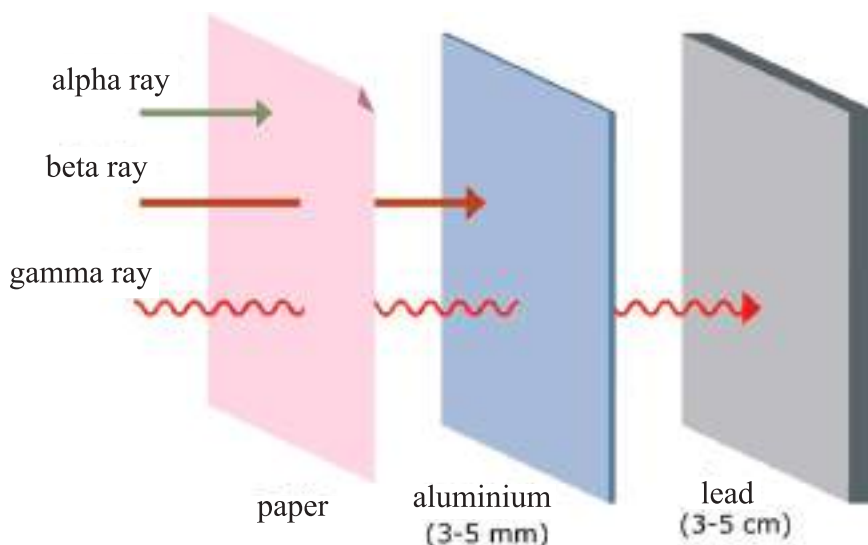
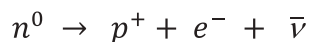


Figure 13.01: It is possible to stop an alpha particle by a single sheet of paper, because it can lose its energy by too much ionizing. It is necessary to stop beta ray or electrons using an aluminum sheet of a few millimeter thick. A thick lead sheet is needed to stop gamma ray.

13.1.2 Beta ray

A Beta particle is actually an electron (it is of course a matter of wonder that a nucleus consists of neutrons and protons only, so how does an electron come out from a nucleus?) For this, inside the nucleus a neutron has to dissociate into a proton and an electron.

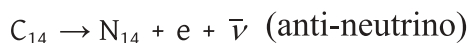


That is, a charge-less neutron is converted into a positively charged proton and a negatively charged electron, so the total amount of charge remains unchanged. In the right side of the equation, ($\bar{\nu}$) indicates the anti particle of a neutrino (anti-neutrino), a mysterious particle of the world of physics, which is chargeless and its mass is very small.

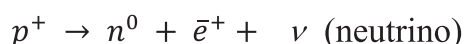
When an alpha particle is emitted from a nucleus, it comes out with a definite amount of energy. But in the case of a beta particle, this is not true. The energy of the beta particle depends on the part of the total energy radiation received by the anti-neutrino.

Since a beta particle is negatively charged electron, it can be influenced by electric and magnetic field. When it passes through matter, it can ionize the molecule-atoms of the matter due to collision with them. Since the size of a beta particle is much smaller than that of an alpha particle (helium nucleus), its penetrating capability is more than the alpha particle and it can penetrate deeper into matter. It is possible to stop a normal beta particle by a few millimeter thick aluminum sheet.

When radiation of beta particles occurs, a proton is increased in the nucleus by deducting of a neutron. So the number of nucleons remains the same. As the atomic number depends on the number of protons, the atomic number of the nucleus increases, and hence a different nucleus of a different element is formed. For example: radioactive C_{14} is converted into N_{14} due to beta radiation:



An interesting matter is that, not only the emission of electrons is beta radiation from inside the nucleus; the emission of an anti-particle of electron (positron) is also called beta radiation. For this reason, a proton has to be converted into the neutron inside the nucleus:



(This reaction cannot take place outside the nucleus. This is because the mass of a neutron is more than that of a proton.)

In this process, the number of protons decreases by one and hence the atomic number also decreases by one, and a nucleus of different element is formed.

Though neutrinos or anti-neutrinos are emitted during beta radiation, they are not considered to be radioactive rays. This is because they are charge less and they are so inert (or nonreactive) that even a few light years thick lead sheet cannot stop a neutrino particle.

13.1.3 Gamma ray

A gamma ray is actually an energetic electromagnetic wave. It has no charge, but as it is strong, its wavelength is very low (frequency is very high). Regardless of the energy of the gamma ray, its velocity is equal to that of light. When a nucleus

becomes excited by radiating alpha or beta particle then extra energy comes out as gamma ray and the nucleus de-excites. A gamma ray is charge less and mass less, so the atomic number or nucleon number of a nucleus does not change due to this radiation.

Since gamma ray doesn't have any charge, so it cannot be influenced by an electric or magnetic field. Because it is charge less, it cannot directly ionize atoms or molecules, but the electrons can ionize atoms or molecules in different processes and from that we can understand the presence of gamma ray. To stop gamma rays with energy equal to that of alpha and beta particles, a thick layer of lead of several centimeters is needed.

13.1.4 Half Life

It is not possible to say the exact moment when a definite radioactive nucleus will emit radioactive rays. Physics can only predict the probability of its radiation. So the concept of Half Life is used to determine the amount of radioactivity. The time in which half of the nucleus disintegrates (or radiates) is known as half life. Therefore, the more the radioactivity of an element, the less its half life is. The half life of a stable nucleus having no radioactivity can be considered to be “infinity”.

It is necessary to know that radioactivity is a nuclear phenomenon, so a nucleus is transformed into another through radioactive emission. A different nucleus can accept or donate one or two electrons from the surroundings to become a charge less atom. This is because the energy of the electrons of an atom is much smaller than the nuclear energy of a nucleus.



Example

Question: The half life of a radioactive element of 1 kg is 100 years. What will be its mass after 200 years?

Answer: Mass does not change directly due to radioactivity. Only three fourth of the nuclei of the radioactive element will emit radioactive particles in 200 years.

13.1.5 Uses of Radioactivity

There are many different uses of Radioactivity. If a small amount of radioactive sample is inserted into the body, information about the body can be detected by observing the movement of the radioactive sample from outside the body. Generally those radioactivity samples have a very low half life, it may a few minutes, thus the radioactivity of the samples will be finished within an hour.

Another important use of radioactivity is the determination of the age of ancient fossils. In our body there is huge amount of carbon in which a definite amount of C_{14} is present. When an animal dies, no more C_{14} can enter the body further. The amount of carbon deposited before gets less due to the half life. Thus knowing the amount of C_{14} that would have normally been present initially, and by calculating the amount which is decayed, age can be determined accurately.

A radioactive element can cause harm to the cells of the body, so different types of caution are taken. Radioactivity is used to destroy harmful cells within the body. This is why, for the treatment of cancer radioactive elements are used to destroy cancer cells.

Radioactivity is also used to make the equipment germ free, detecting the presence of smoke and determining the amount of different metals in minerals.

13.1.6 Cautions about Radioactivity

A high radioactive dose can cause many problems in our body. When research started about radioactivity, the scientists did not know the bad effects of radioactivity. As a result, scientists were attacked with many diseases as they came close contact with radioactivity. Due to working for a long time with radioactive elements Marie Curie suffered from leukemia and died. Radioactivity decreases immunity against disease, preventing capability of human and even physical disable children are born inherently.

We do not usually face radioactivity often in our daily life. Due to new



Figure 13.02: Radioactivity warning sign

technology in the world, many people have started to face the bad effects of radioactivity. The nuclear reactions in a nuclear power plant produce serious radioactivity. Many radioactive wastes have high half life and can remain radioactive for lakhs of years. There are even incidents of radioactive wastes being spread into the surroundings due to accidents in nuclear power plants. Many people faced radioactivity in the accident of a submarine run by nuclear energy. The most devastating event took place in Hiroshima and Nagasaki due to nuclear bombs. Many people were exposed to radioactivity. So, many researchers have been started becoming concerned about radioactivity and the determination of the safe limit of radioactivity is also started. General people are now being warned about the effects of radioactivity (figure 13.02).

13.2 Development of Electronics

Electronics has a greatest contribution in the development of the modern civilization and it is not an exaggeration by any means. The gradual development of electronics can be divided into three parts, the vacuum tube, transistor and integrated circuit.

13.2.1 Vacuum Tube

In 1883 Edison observed that in a light bulb, a current can flow through a vacuum space from the filament to a metal plate. This process is known as the Edison Effect. In 1904, John Fleming (on the basis of the Edison effect) first created a vacuum tube with two electrodes, which worked as a rectifier i.e., it converted an alternating current into a direct current. This vacuum tube is considered as the introduction to electronics. At that time the exchange of information was started by radio waves. Guglielmo Marconi needed such type of vacuum tube to control of the radio waves. (It is noted that Marconi was mentioned as the inventor of radio, but in recent times the contribution of Bengali scientist Jagadish Chandra Bose is also acknowledged.)

In 1906 Lee De Forest invented another vacuum tube by adding a third electrode, which is known as triode. This triode can control the flow of electricity and used as an amplifier. The development of vacuum tubes take place first through telegraph



Figure 13.03 : Types of vacuum tube

communication using Morse code and then through the use of vocal communication by telephone (Figure 13.03). During the first and second world war, different types of specialized vacuum tubes were used in controlling weapons, radar, navigation etc. In 1946 the first computer ENIAC was invented using 1800 vacuum tubes.

13.2.2 Transistor

In 1947 the Transistor was first invented in Bell Laboratory and for this invention John Bardeen, Walter Brattain and William Shockley were awarded the Nobel Prize. No one could imagine at that time how fast and widely the transistor would change the whole world.

A transistor can work just like a vacuum tube, but compared with a vacuum tube it is very small in size, light in weight, and a very small amount of electricity is needed to operate it, it is very reliable and most of all, it can be made at a very low cost. Thus it rapidly secured its place, replacing the vacuum tube and the people of the world began to get many electrical appliances by using this transistor at very cheap.

13.2.3 Integrated Circuit

Although the discussion about an integrated circuit was started in 1952, a real integrated circuit was started to be made in the sixth decade. In the fifth decade many transistors were made on a thin silicon plate (Wafer) and eventually the transistors were separated by cutting them from the plate. This process was improved to a small extent while preparing an integrated circuit. Not only a transistor but also creating a diode, resistor and a capacitor on a wafer, a complete circuit was started to be made. This circuit was named an integrated circuit (IC). With the development of technology many transistors were assembled in a small area, and first it was named Large Scale Integration (LSI), and then it is called Very Large Scale Integration (VLSI). These circuits were made into a ready-made package so that they could be directly used in a circuit board. Modern equipment such as the Micro-computer, clinical appliances, video camera, communication satellite would never be possible without the use of integrated circuits.

13.2.4 Electronics of Future

The technology of electronics is still advancing, and in the future we will

experience electronic circuits using optics or light in an information communicative integrated circuit. At the same time we will observe the use of programmable IC (FPGA: Field Programmable Gate Array) in different fields in accordance to our own needs.

13.3 Analog and Digital Electronics

Things that are happening around us every moment, like sound, light, pressure, temperature are expressed as information or data. Their values can be changed continuously. We might need these values in our different works, so we preserve and analyze these values and send them from one place to another. To process the data we might use electronics. This continuously changing data can be converted into electrical signals and these types of signals are called analog signals. If we process these signals with a type of electronics, it is called analog electronics. When discrete signals are processed, then it is called digital signal.

Continuously changing information or data signals can be processed in a different way. To do this, we have to find the value for a definite interval of time and express this in a type of number. Then we have to preserve this value successively. Then we will be able to process these numbers using electronics. When it is needed to convert this into the main analog signal again, then electric signals of equal to serially preserved values have to be created. We use decimal numbers in our daily life. But in electronics numbers are expressed by binary numbers, because they can be easily processed (voltage as 1 and zero voltage as 0). The number system expressed with the digits '0' and '1' is a binary number system. (Figure. 13.04)

The greatest contribution of electronics is the computer. The exchange of all data or processing of all data in a computer is processed by digital electronics. Digital electronics is also used in the internet and computer networks to exchange data. Sound, image or video etc. start as analog signals (and is used as analog signal too) but they are stored, processed and sent as digital signal. Noise can spoil the quantity of analog signals very easily by entering the signals. Once it is converted into digital signals, noise cannot enter it very easily. So, the quality of digital signals remains unchanged.

A special type of IC is made to process digital signals. These IC's have become efficient gradually. That is, they can process a large number of signals at a very short time accurately. Therefore, as time goes by, the process of digitizing is becoming easier and it is needless to say that everything around us is being converted into digital world.

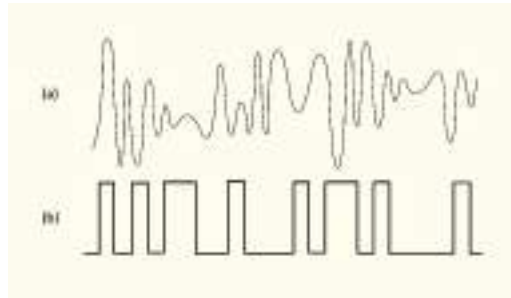


Figure 13.04: (a) Analog (b) Digital signals

13.4 Semiconductor

The modern world and modern civilization have been built on the basis of electronics. If we want to express our gratitude to any substance, that substance will be the semiconductor. We have discussed about the conductor, insulator and semiconductor. Now we can go deeper to a small extent about this. In Figure 13.05 many atoms of a

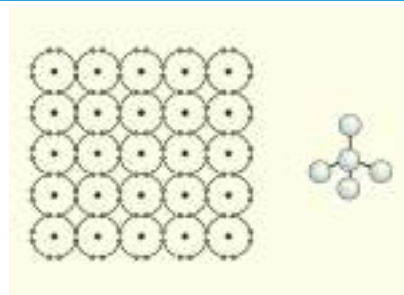


Figure 13.05 : Silicon crystal.

Three dimensional silicon crystal (right)

Due to the structure of the atom if there are eight electrons in their outer most shells, in one sense they are fulfilled and become very stable. Atoms always try to have eight electrons in their outer most shells. Silicon is the most used semiconductor among the semiconductors. The number of electrons is four in its outer most shell. But when we observe a silicon crystal, we find surprisingly that there are eight electrons in the outer most shell of every atom! It has happened because every atom attaches to four different atoms around it, and they are all using their electrons sharing with the neighboring atoms. (We have drawn the figure on a plane. The real arrangement of silicon atoms is three dimensional, as shown at the right side of the Figure 13.05. Actually, every atom touches other four atoms.)

Usually the electrons are bound to the atoms in semiconductors. With the increase of temperature, one or two electrons may become free. There are free electrons in conductors so the semiconductors may act as conductors to some extent. From the view point of physics it is an interesting phenomenon, but not very suitable for practical use. An interesting thing is done to make it more useful. Atoms with five electrons in its outer most shell. e.g, phosphorous are mixed with silicon crystal. Hence, the fifth

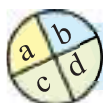
electron of phosphorous becomes an extra electron. It is not necessary for the extra electron to be bound to the atoms of phosphorus. This electron behaves like free electrons by making the phosphorus atom a positive ion. It can be said that such phosphorus adding semiconductors act as conductors to a large extent, because there are some free electrons here to carry charges. The semiconductors made by mixing atoms having five electrons in the outer most shell like phosphorus are called n type semiconductors.

Now get ready to hear something more interesting. Instead of adding pentavalent impurity atoms, if we add atoms having three electrons (e.g. boron) in their outer most shells, to silicon, what will happen? Certainly, it is understood that there is an empty space in the orbit of a boron atom and this empty place can be occupied by a neighboring electron. Then an empty place will be created in the neighboring atom, this place again can be occupied by an electron of its neighboring atom, then an empty place also will be created there. Another way to see it is that an empty place with the lack of an electron is moving from atom to atom. It seems that it is one type of particle and its charge is positive. It is called a hole. That is, we can say that keeping a boron atom as a negative ion, this hole can move through the whole silicon crystal. That is, this semiconductor almost acts as a conductor and positively charged holes conduct electricity through it! When a semiconductor becomes a conductor by adding atoms having three electrons in their outer most shells, it is called a p-type semiconductor.

There were not so many uses of n type and p type semiconductors individually but when n type and p type semiconductors were combined together the greatest discovery of science and technology started.



Sample Questions



Multiple choice questions

Put the tick mark (✓) on correct answer

1. What do you mean by an alpha particle emitting from a radioactive element?
 - (a) a hydrogen nucleus
 - (b) a helium nucleus
 - (c) a neutrally charged particle
 - (d) a negative particle
2. What is a beta ray emitted due to the radioactive decay?
 - (a) flow of negative electrons
 - (b) a charge neutral particle
 - (c) a light nucleus
 - (d) flow of positive protons

3. What is a silicon chip called if millions of circuits are added to it?

- (a) parallel circuit
- (b) semiconductor transistor
- (c) integrated circuit
- (d) semiconductor diode

4. Integrated Circuit—

- i. is easy to use
- ii. is made of semiconductor material
- iii. is used to reduce the weight of electronic devices

Which one is correct?

- (a) i
- (b) ii
- (c) ii & iii
- (d) i, ii & iii



Creative questions

1. Tanmoy was researching a radioactive element in the physics lab. A 1 kg sample of the element was stored, and after 900 years, its radioactive atomic mass was found to be 250 gm. During his research, Tanmoy observed a ray or particle emitted from the radioactive element that had the same mass as an electron and a negative charge. He also detected another ray that travels at the speed of light and is not affected by magnetic fields.

- (a) What is a binary number?
- (b) Using a transistor is more advantageous than a vacuum tube- Explain.
- (c) Calculate the half-life of the radioactive element.
- (d) Identify the rays mentioned in the stem and compare their characteristics.



Short Answer Questions

- 1. Explain what is meant by radioactivity.
- 2. Explain the difference between alpha and beta particles.
- 3. What happens to the resistance of a semiconductor when the temperature increases? Explain
- 4. What is meant by the half-life of a radioactive element?

---End---